REGULARITY DESCENDS ALONG A FLAT FINITELY PRESENTED SURJECTION

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Allowed standard prerequisites (editorial).

- Noetherianness descends along a faithfully flat finitely presented ring map. Algebra, Lemma algebra-lemma-descent-Noetherian, invoked at proof lines 5316-5318. Gives that A is Noetherian.
- Regularity of the source of a flat ring map forces regularity of the target. Algebra, Lemma algebra-lemma-flat-under-regular, invoked at proof lines 5319-5320. Gives that A is regular.

Source reference. Original lemma 033E: sources/stacks-chow/tags.txt, source lines 4053–4053. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00OF: sources/stacks-chow/tags.txt, source lines 893–893. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of algebraic spaces. If f is locally of finite presentation, flat, and surjective and X is regular, then Y is regular.*

Proof. This lemma reduces to the following algebra statement: If $A \to B$ is a faithfully flat, finitely presented ring homomorphism with B Noetherian and regular, then A is Noetherian and regular. We see that A is Noetherian by Algebra, Lemma 033E and regular by Algebra, Lemma 00OF. □

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